```
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
```

```
# Load the MNIST dataset
(x_train, y_train), (x_test, y_test) = mnist.load_data()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
# Print the shape of the data
print("Training data shape:", x_train.shape)
print("Test data shape:", x_test.shape)
```

```
Training data shape: (60000, 28, 28)
Test data shape: (10000, 28, 28)
```

```
# Reshape to add channel dimension (grayscale = 1 channel)
x_train = x_train.reshape((x_train.shape[0], 28, 28, 1))
x_test = x_test.reshape((x_test.shape[0], 28, 28, 1))
```

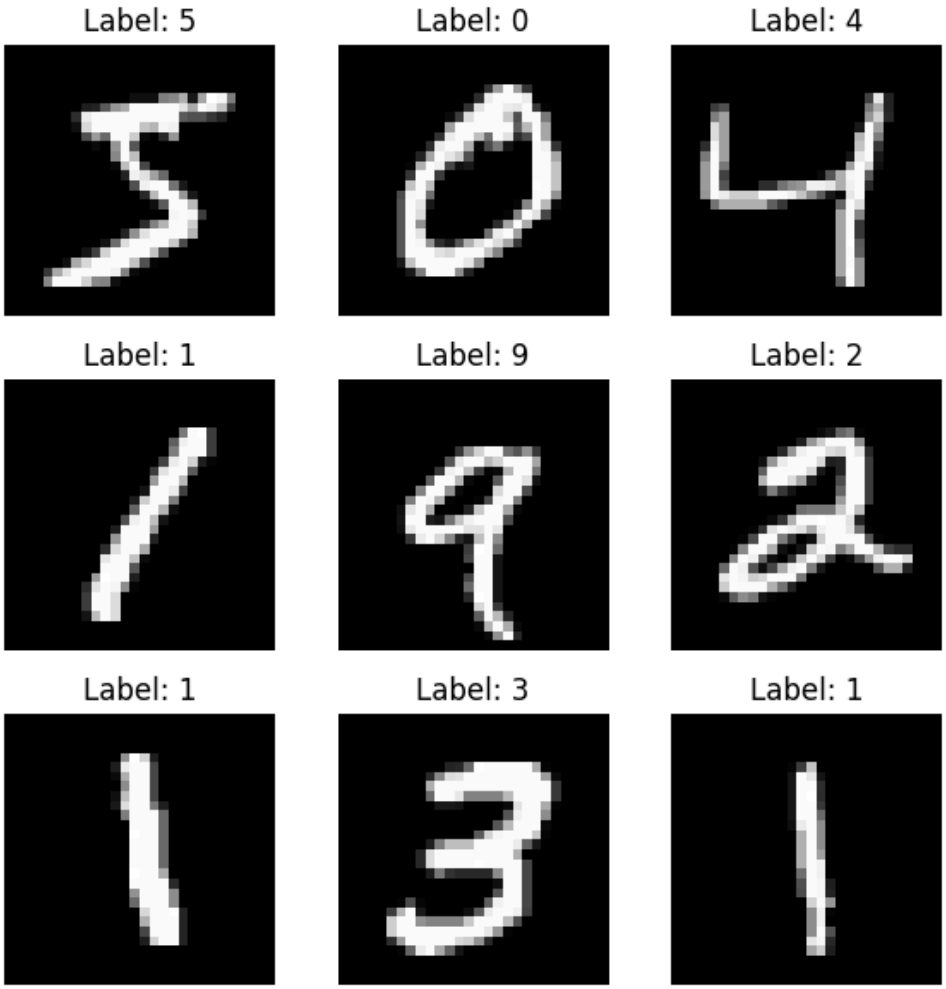
```
# Normalize pixel values
x_train = x_train.astype("float32") / 255
x_test = x_test.astype("float32") / 255
```

```
# One-hot encode the labels
y_train = to_categorical(y_train)
y_test = to_categorical(y_test)
```

```
# Check the shape
print("x_train shape:", x_train.shape)
print("y_train shape:", y_train.shape)
```

```
x_train shape: (60000, 28, 28, 1)
y_train shape: (60000, 10)
```

```
# Visualize the first 9 digits
plt.figure(figsize=(6, 6))
for i in range(9):
 plt.subplot(3, 3, i + 1)
 plt.imshow(x_train[i].reshape(28, 28), cmap='gray')
 plt.title(f"Label: {np.argmax(y_train[i])}")
 plt.axis('off')
plt.tight_layout()
plt.show()
```




```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (
    Conv2D, MaxPooling2D, Flatten,
    Dense, Dropout
)

# Define the CNN model
model = Sequential([
    Conv2D(32, kernel_size=(3, 3), activation='relu', input_shape=(28, 28, 1)),
    MaxPooling2D(pool_size=(2, 2)),
    Conv2D(64, kernel_size=(3, 3), activation='relu'),
    MaxPooling2D(pool_size=(2, 2)),
    Flatten(),
    Dense(128, activation='relu'),
    Dropout(0.5),  # reduce overfitting
    Dense(10, activation='softmax')  # 10 output classes
])
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113:
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Input, Conv2D, MaxPooling2D, Flatten, Dense

model = Sequential([
    Input(shape=(28, 28, 1)),   # 👈 add this
```

```
    Conv2D(32, (3,3), activation='relu'),
    MaxPooling2D((2,2)),

    Conv2D(64, (3,3), activation='relu'),
    MaxPooling2D((2,2)),

    Flatten(),
    Dense(128, activation='relu'),
    Dense(10, activation='softmax')
])
```

```
model.compile(optimizer='adam',loss='categorical_crossentropy',metrics
=['accuracy'])
```

```
# Train the model
history = model.fit(
 x_train, y_train,
 epochs=5,batch_size=128,
 validation_split=0.1,
 verbose=2
)
```

```
Epoch 1/5
422/422 - 47s - 111ms/step - accuracy: 0.9344 - loss: 0.2271 - val_accuracy: 0.9733 - va
Epoch 2/5
422/422 - 83s - 197ms/step - accuracy: 0.9811 - loss: 0.0618 - val_accuracy: 0.9860 - va
Epoch 3/5
422/422 - 98s - 231ms/step - accuracy: 0.9866 - loss: 0.0437 - val_accuracy: 0.9857 - va
Epoch 4/5
422/422 - 54s - 128ms/step - accuracy: 0.9896 - loss: 0.0328 - val_accuracy: 0.9890 - va
Epoch 5/5
422/422 - 55s - 129ms/step - accuracy: 0.9920 - loss: 0.0259 - val_accuracy: 0.9870 - va
```

```
test_loss, test_acc = model.evaluate(x_test, y_test)
print(f"Test Accuracy: {test_acc:.4f}")
print(f"Test Loss: {test_loss:.4f}")
```

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step - accuracy: 0.9876 - loss: 0.0390
Test Accuracy: 0.9876
Test Loss: 0.0390
```

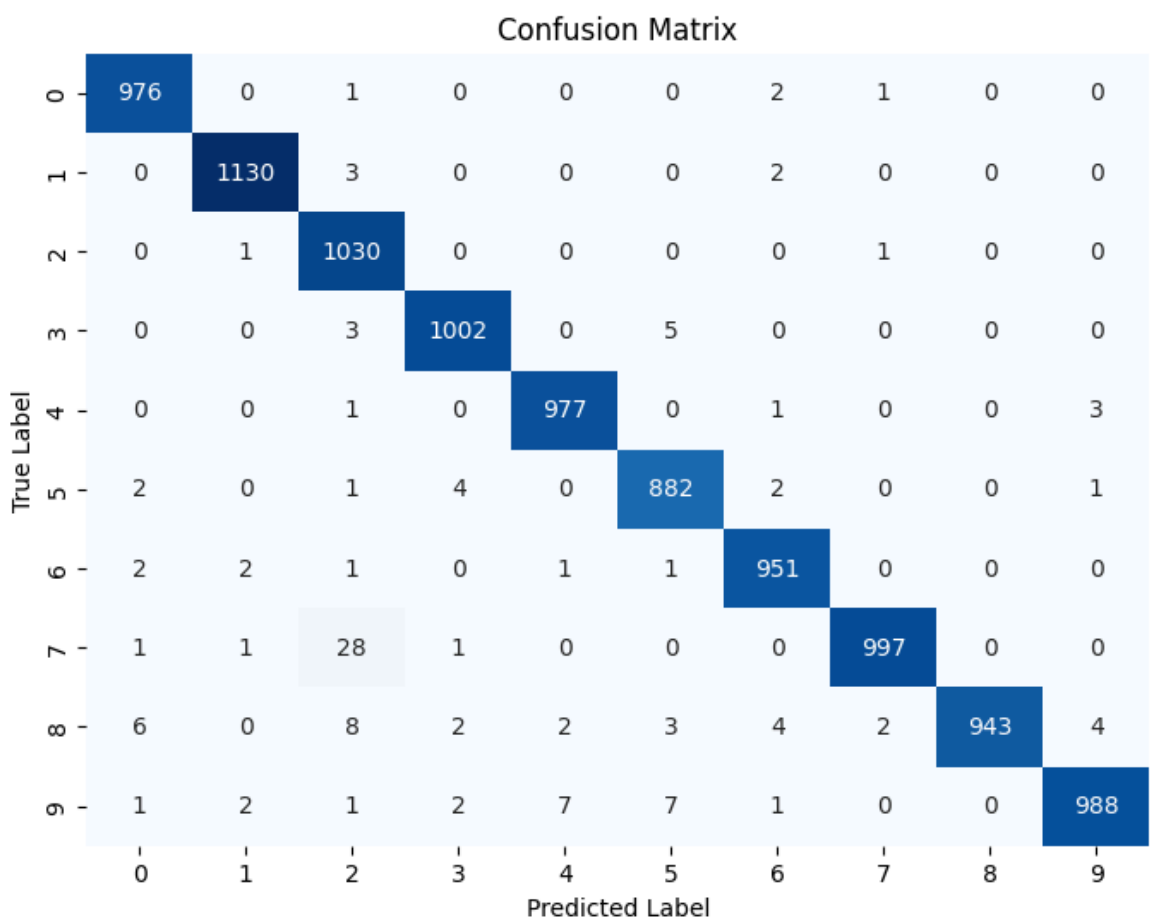
```
from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns
# Predict the classes
y_pred = model.predict(x_test)
y_pred_classes = np.argmax(y_pred, axis=1)
y_true = np.argmax(y_test, axis=1)
# Classification report
print("Classification Report:\n")
print(classification_report(y_true, y_pred_classes))
```

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 10ms/step
Classification Report:

              precision    recall  f1-score   support

           0       0.99      1.00      0.99       980
           1       0.99      1.00      1.00      1135
```

```
              2       0.96      1.00      0.98      1032
              3       0.99      0.99      0.99      1010
              4       0.99      0.99      0.99       982
              5       0.98      0.99      0.99       892
              6       0.99      0.99      0.99       958
              7       1.00      0.97      0.98      1028
              8       1.00      0.97      0.98       974
              9       0.99      0.98      0.99      1009

    accuracy                           0.99     10000
   macro avg       0.99      0.99      0.99     10000
weighted avg       0.99      0.99      0.99     10000
```

```
# Confusion matrix
conf_matrix = confusion_matrix(y_true, y_pred_classes)
```

```
# Plot the confusion matrix
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues',
cbar=False)
plt.title("Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.show()
```

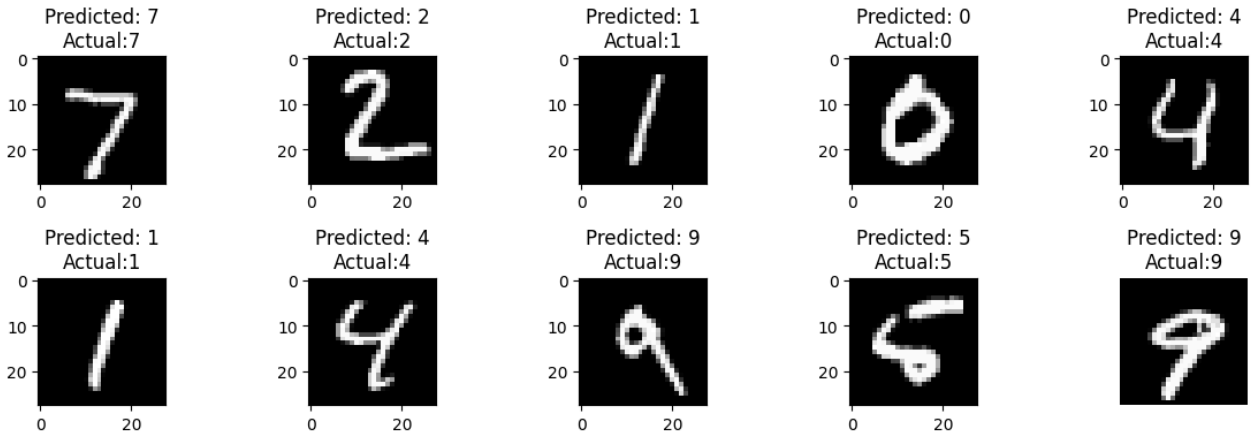



```
# Predict class probabilities for test images
predictions = model.predict(x_test)
```

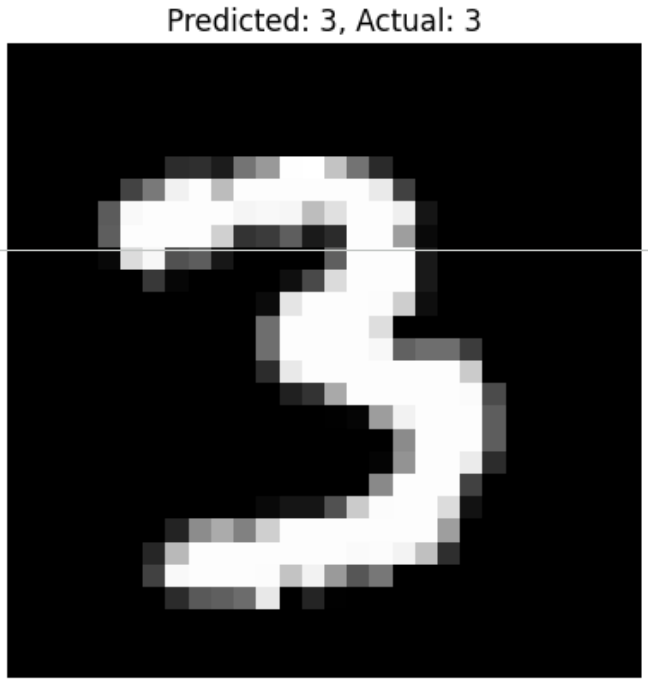
```
313/313 ━━━━━━━━━━━━━━━━━━━━ 4s 13ms/step
```

```
# Convert probabilities to class labels
predicted_classes = np.argmax(predictions, axis=1)
true_classes = np.argmax(y_test, axis=1)
```

```
# Plot 10 test images with predictions
plt.figure(figsize=(12, 4))
for i in range(10):
 plt.subplot(2, 5, i + 1)
 plt.imshow(x_test[i].reshape(28, 28), cmap='gray')
 plt.title(f"Predicted: {predicted_classes[i]}\nActual:{true_classes[i]}")
plt.axis('off')
plt.tight_layout()
plt.show()
```




```
# Predict a single image (e.g., index 500)
index = 500
single_image = x_test[index].reshape(1, 28, 28, 1)
```

```
# Predict and show result
predicted_label = np.argmax(model.predict(single_image))
actual_label = np.argmax(y_test[index])
plt.imshow(x_test[index].reshape(28, 28), cmap='gray')
plt.title(f"Predicted: {predicted_label}, Actual: {actual_label}")
plt.axis('off')
plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step
```